Problem-Solving Insights IV

Estimations

Lexington Physics Club

Estimation is an essential skill for a physicist. It could quickly check your answers or give you wonderful insights for complicated scenarios. The qualitative thinking behind estimations is also the bridge that connects your physics knowledge to real life.

In fact, some of the hardest IPhO and EuPhO problems are straight forward estimation problems! That's because many seemingly impossible setups can be analyzed without being able to find the exact answers.

Process of estimating:

Variable identification: Define the quantity to be estimated, the relevant variables, and the system boundary. *Dimensional analysis:* Use units to constrain possible relationships and check consistency. *Special cases:* Examine limiting or simplified cases to infer dependencies and test a proposed relationship. *Derive an expression:* State the assumptions and derive relations; dimensional analysis may fail when there are variables of the same dimensions. *Look for values:* Measure or research the inputs, recording sources and plausible ranges. *Calculate:* Keep track of appropriate significant figures, and identify the dominant uncertainty.

1. The cost of charging a phone

Estimate the annual electricity cost of charging a phone. Then estimate the cost if every student at LHS charged a phone once on each school day.

2. Carbon dioxide in a classroom

Estimate how much the carbon dioxide concentration could rise during one class period if outdoor-air exchange were negligible. Then estimate the outdoor-air flow needed to maintain a given indoor–outdoor concentration difference.

3. Vinegar and an eggshell

Estimate the minimum volume of household vinegar required to dissolve the mineral portion of one eggshell, and the amount of carbon dioxide that the reaction could produce. Treat the reactive mineral as calcium carbonate and the acid as acetic acid.

4. A classroom's genomes on a phone

Estimate the storage required to represent a human genome as a sequence of DNA letters. Could the available storage on your phone hold one such sequence for every student in your class? For every student in your school?

5. Temperature and musical pitch

An organ is tuned in a cool room before a concert. Estimate how much its pitch changes when the room warms. Decide whether the change is significant compared with a musical semitone.

(a) Choose a plausible initial temperature, temperature increase, and note frequency. Research the relation between sound speed and air temperature, and the frequency ratio corresponding to a semitone. Use absolute temperature where required.

(b) For the same resonance mode, model the frequency as proportional to sound speed divided by pipe length. Initially neglect pipe expansion. Derive the frequency ratio after warming, and check that it tends to one when the temperature change vanishes.

(c) Calculate the shift in hertz and as a fraction of a semitone. You might want to express it in cents, with 100 cents per semitone. Compare two starting notes: does the same warming produce the same hertz shift or the same fractional shift?

(d) Research a plausible pipe material and estimate whether thermal expansion changes the conclusion. Explain why the sound-speed argument does not directly predict the pitch change of a guitar string, and identify the relevant quantities for a string model.

6. Jumping off an asteroid

Estimate the largest asteroid from which an astronaut could escape with a standing jump. Model the asteroid as an isolated, nonrotating sphere of uniform density, neglect other bodies, and assume the astronaut can leave the surface with the same speed inferred from an Earth-based vertical jump.

(a) Choose a plausible Earth jump height, measured as the rise of the center of mass after takeoff. Use energy conservation to infer the takeoff speed.

(b) Express the asteroid's mass in terms of radius and density. Derive its escape speed, then obtain the limiting radius by comparing escape speed with the assumed takeoff speed.

(c) Research plausible asteroid densities and calculate a range of limiting radii.

(d) Determine how the limiting radius scales with jump height and asteroid density. Estimate the surface gravity at your limiting radius. Discuss the potential effects caused by rotation, irregular shape, loose surface material, and a spacesuit.